جامعة الأخوين
ⵜⴰⵙⴷⴰⵡⵉⵜ ⵏ ⵓⵅⴰⵡⵉⵏ
AL AKHAWAYN UNIVERSITY



A COMPARATIVE STUDY OF DCGAN VS DIFFUSION MODELS FOR IMAGE SYNTHESIS

PRESENTED BY

CHRIFI ALAOUI Khawla
OGBI Noura

SUPERVISED BY

Prof. Asmae MOURHIR

TABLE OF CONTENTS

01 INTRODUCTION

02 GANS OVERVIEW

03 DIFFUSION MODELS

04 OBJECTIVES

05 RELATED WORK

06 MATERIALS & METHODS : DATASET & MODELLING

07 RESULTS & DISCUSSION

08 CONCLUSION

INTRODUCTION

- Image generation is a major task in computer vision and artificial intelligence.
- Generative models aim to learn the distribution of data in order to create new realistic images.
- Among the strongest approaches:
  Generative Adversarial Networks (GANs)
  Diffusion Models
- This project focuses on a comparative study between DCGAN and DDPM for face image synthesis using the CelebA dataset.

GENERATIVE ADVERSARIAL NETWORKS

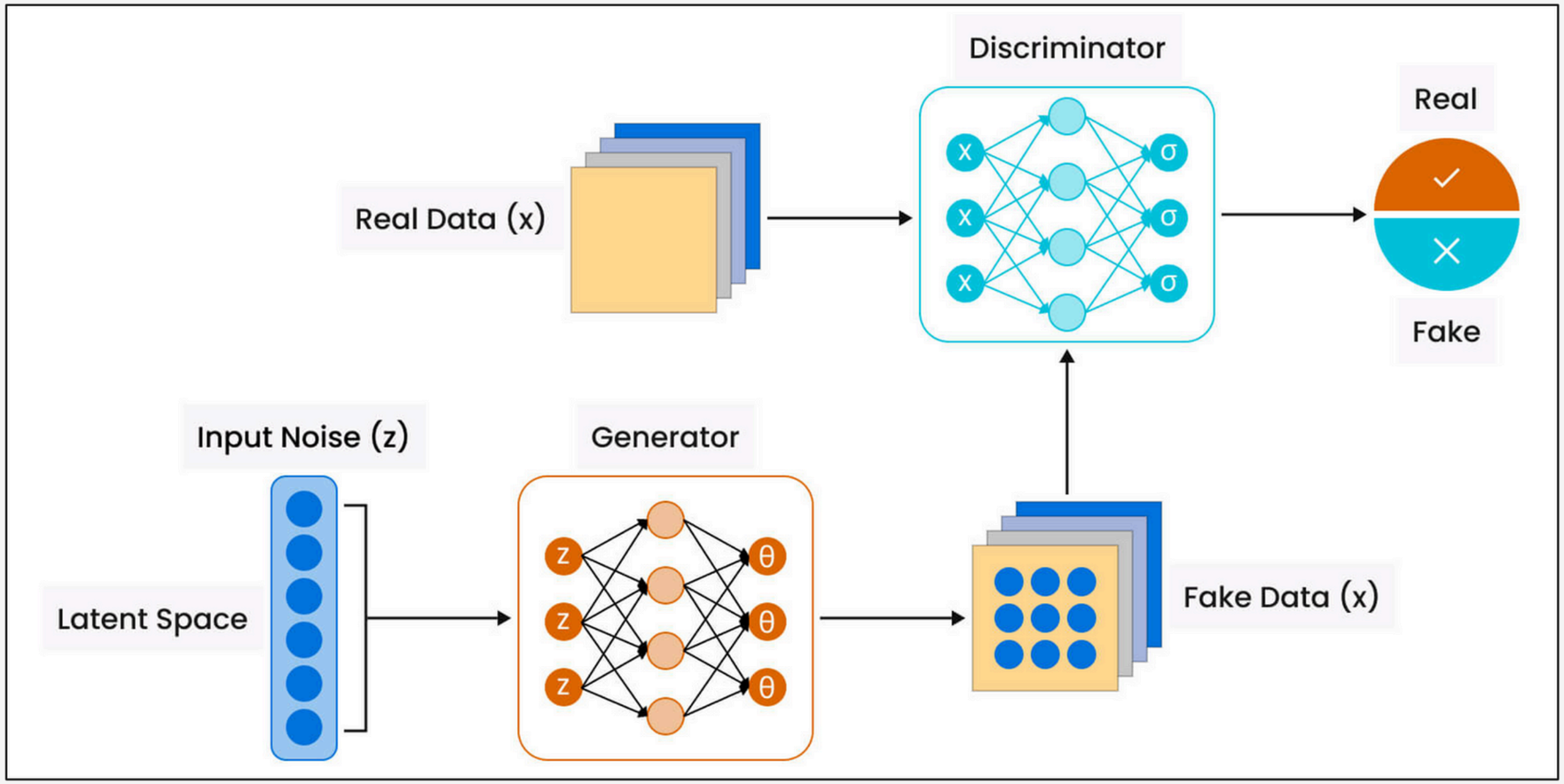
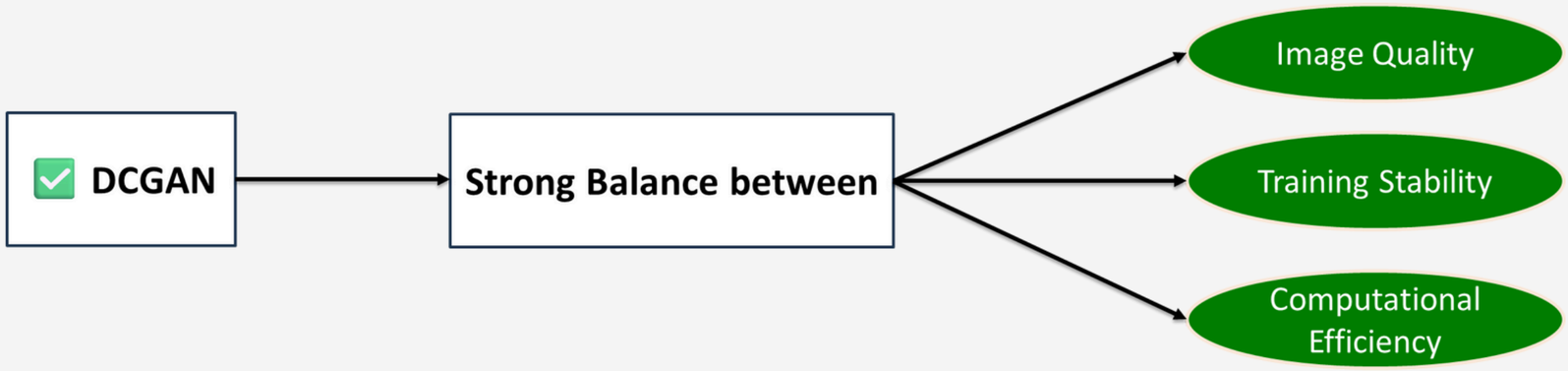


FIGURE 1: GAN ARCHITECTURE

Source

GANS VARIANTS

Model	Description
Vanilla GAN	Cannot model fine facial details, highly unstable, produces blurry and unrealistic faces
CGAN	Can control attributes (e.g., male/female) but still struggles with realism and sharp details
WGAN	Very stable training and better diversity, but not specialized for high-resolution faces
DCGAN	Optimized for images, learns facial features well, produces sharp and realistic faces for CelebA



DCGAN

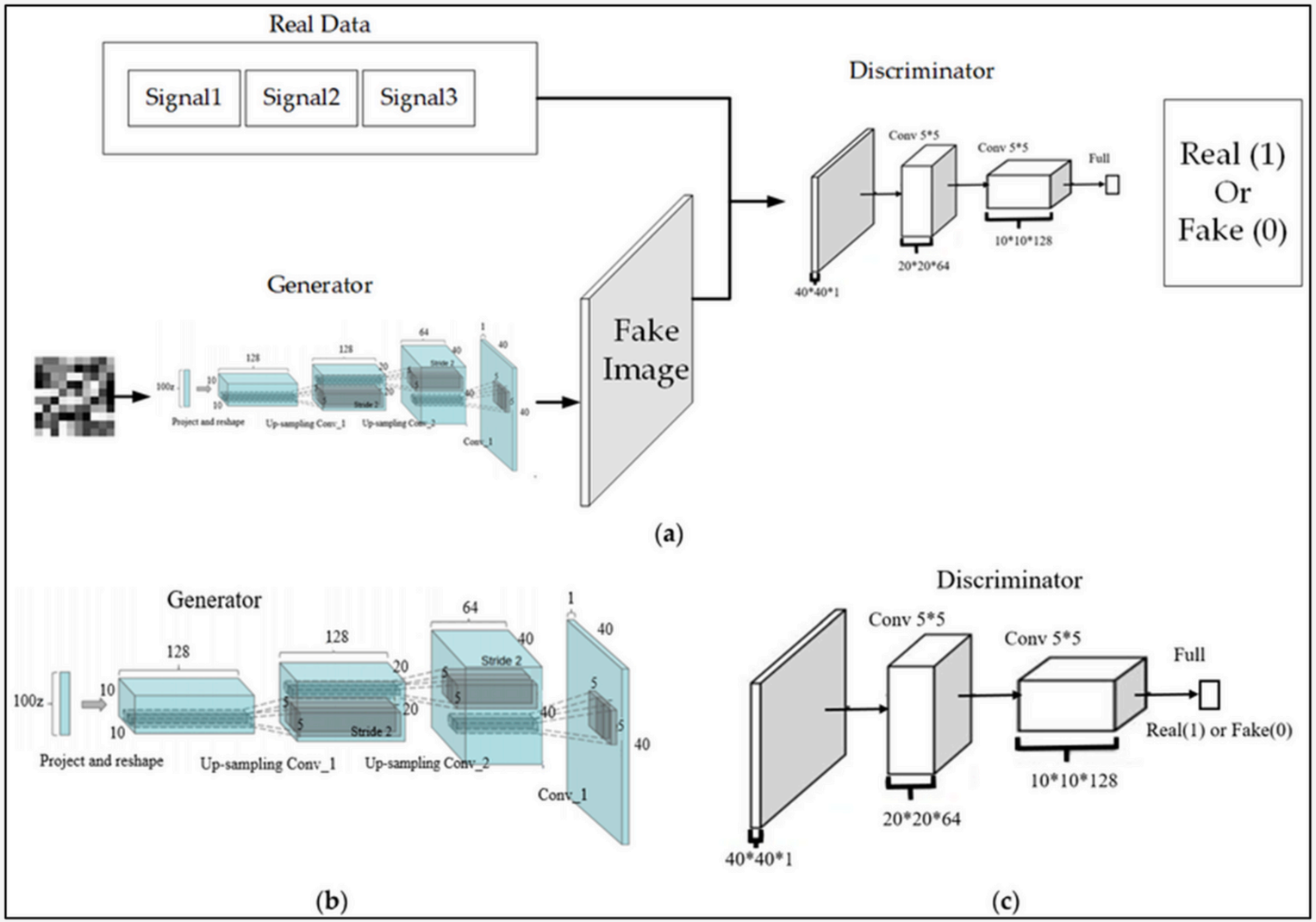


FIGURE 2: DCGAN: (A) DCGAN ARCHITECTURE DIAGRAM; (B) GENERATOR ARCHITECTURE DIAGRAM; (C) DISCRIMINATOR ARCHITECTURE DIAGRAM.

Source

DIFFUSION MODELS

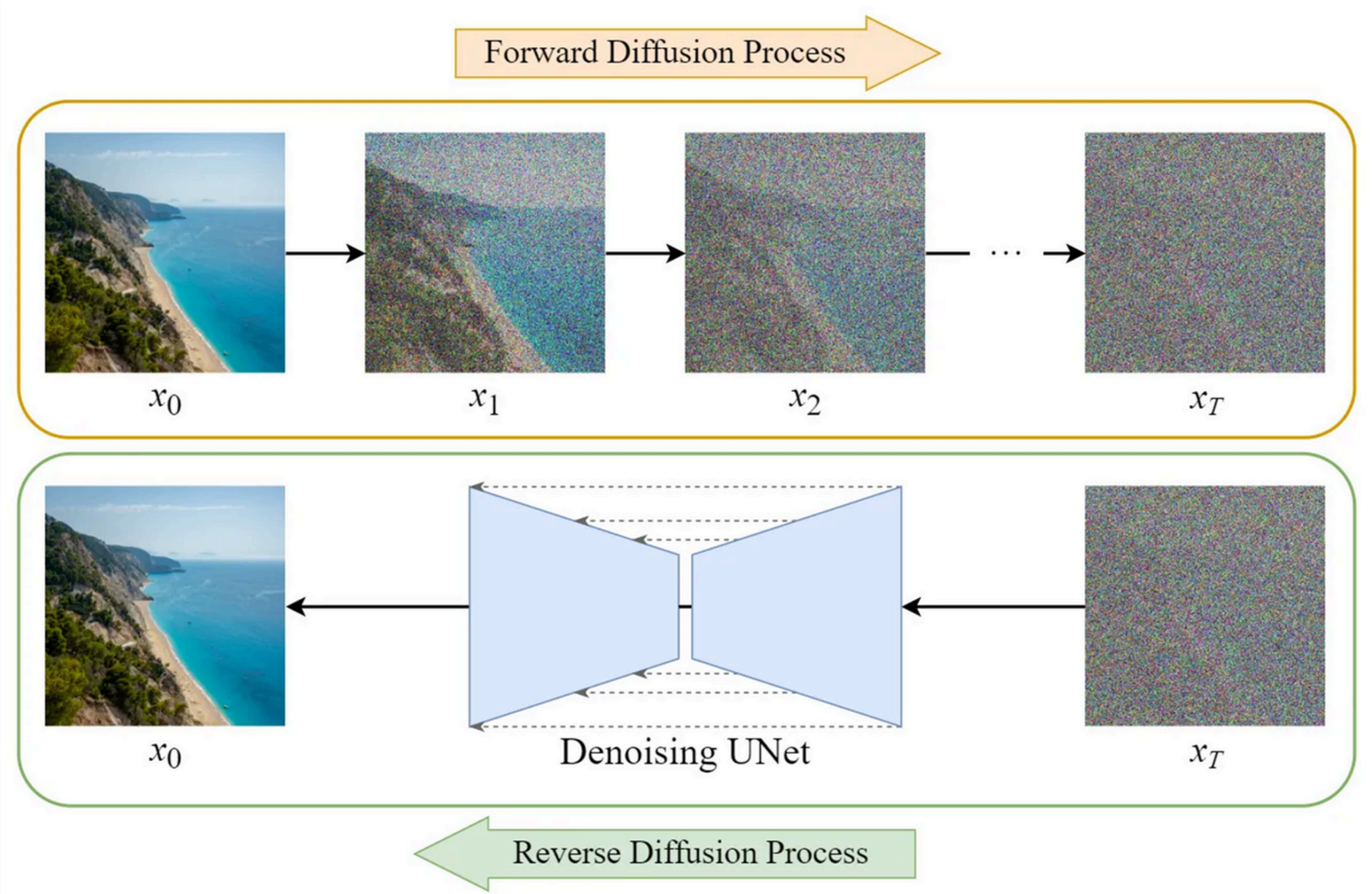


FIGURE 3: DIFFUSION MODEL - FORWARD NOISE PROCESS AND REVERSE DENOISING WITH UNET

Source

DDPM ARCHITECTURE: UNET-BASED NOISE PREDICTION NETWORK

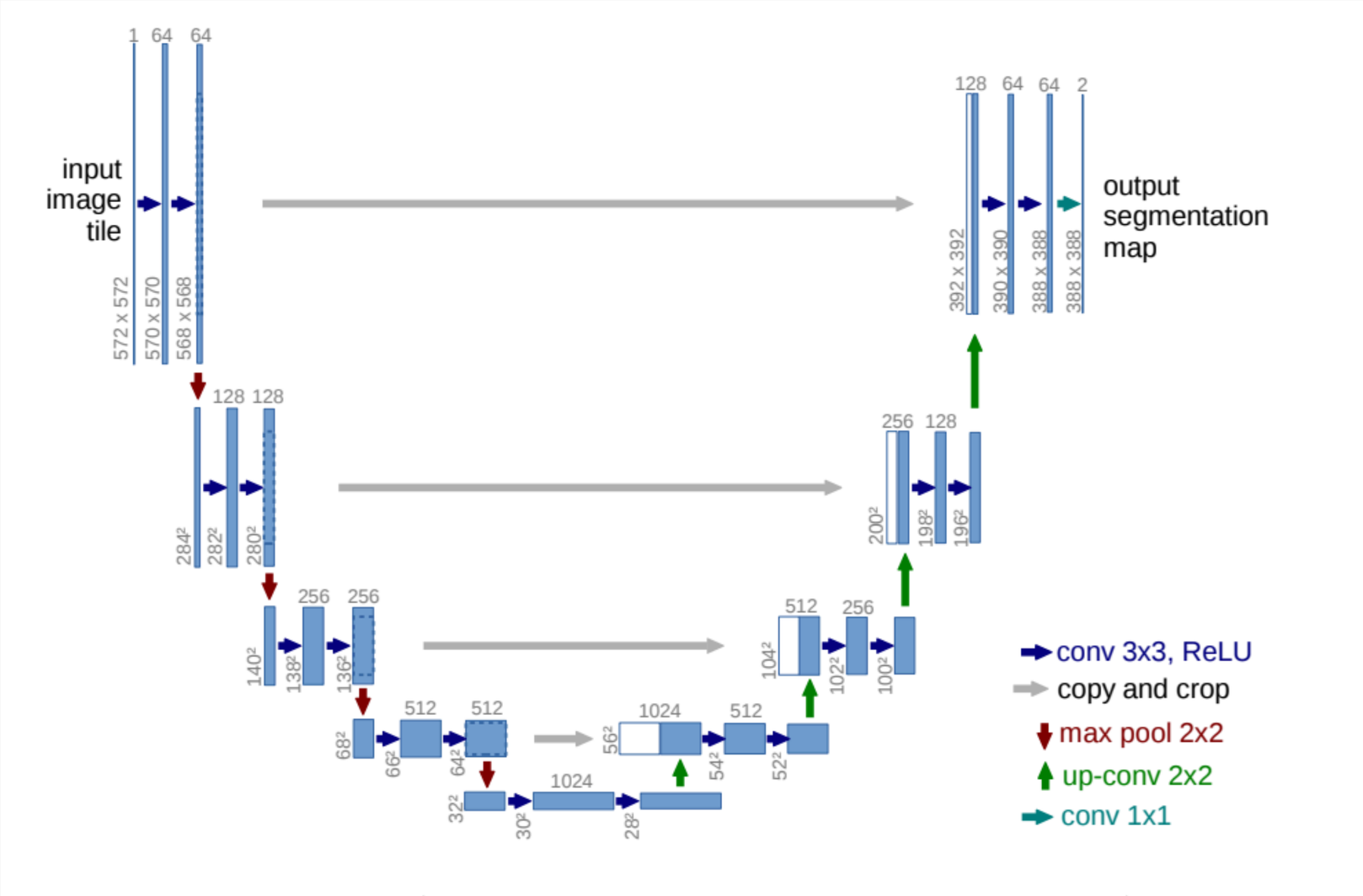


FIGURE 4: UNET-BASED DENOISING NETWORK IN DDPM

Source

OBJECTIVES

“ This study aims to provide a clear understanding of how modern diffusion models outperform classical GAN architectures in image synthesis tasks ”

- Train two generative models: DCGAN and a Diffusion Model.
- Generate realistic face images.
- Measure image quality using FID & IS.
- Compare results and discuss strengths/weaknesses of each model.



Figure 5: Real face images and the fake generated faces by DCGAN

Source

RELATED WORK

Paper	Method	Dataset	Metrics	Key Results
Comparative Analysis of GANs and VAEs (IRJET, 2024)	GAN vs VAE	MNIST	FID, training time	GANs produce sharper images; VAEs better capture latent distribution.
A Primary Comparison of Diffusion Models and GANs for Image Synthesis (2024)	DDPM, Conditional Diffusion vs AC-GAN	MNIST, CIFAR-10	Visual inspection, training stability	Diffusion more stable and better at later stages; GAN trains faster but less stable.
Comparison Between GANs and Diffusion Models for Enhancing Tree Species Recognition (2024)	DCGAN vs Diffusion (U-Net DM) for synthetic data generation	2,178 wood cross-section images (18 species)	FID, IS, F1-score	Diffusion generated more realistic images and improved rare-class classification.
Diffusion-Driven GAN Inversion for Multi-Modal Face Image Generation (2024)	Hybrid Diffusion + GAN (StyleGAN, ControlNet)	Human face datasets (2D & 3D)	Qualitative/visual, reconstruction accuracy	Combining Diffusion + GAN produces more realistic, controllable face images; best of both worlds.

DATASET’S ACQUISITION & PREPARATION

- 202,599 face images, 10,177 identities.
- High diversity.
- Natural variation (age, gender, lighting).



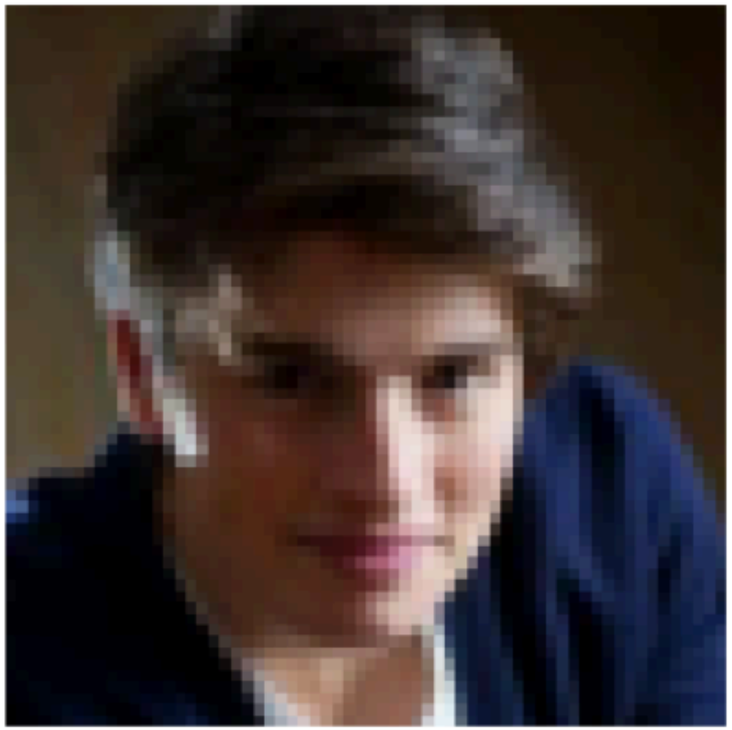
DATA PREPROCESSING

- Crop to center face region.
- Images originally 178×218 (rectangular)--> Resize to 64 × 64.
- Convert to tensors.
- Normalize pixel values → [-1, 1].

Original (CelebA raw)



After preprocessing (64×64)



MODELING APPROACHES

DCGAN GENERATOR (G)

Layer	Details	Output
Input	Latent vector **z = 100**	(100)
Linear	100 → 4×4×512	(512, 4, 4)
BatchNorm + ReLU	—	(512, 4, 4)
ConvTranspose2d	512 → 256, k=4, s=2, p=1	(256, 8, 8)
BN + ReLU	—	(256, 8, 8)
ConvTranspose2d	256 → 128, k=4, s=2, p=1	(128, 16, 16)
BN + ReLU	—	(128, 16, 16)
ConvTranspose2d	128 → 64, k=4, s=2, p=1	(64, 32, 32)
BN + ReLU	—	(64, 32, 32)
ConvTranspose2d	64 → 3, k=4, s=2, p=1	(3, 64, 64)
Tanh	Output image	(3, 64×64)

MODELING APPROACHES

DCGAN DISCRIMINATOR (D)

Layer	Details	Output
Conv2d	3 → 64, k=4, s=2, p=1	(64, 32, 32)
LeakyReLU	—	—
Conv2d	64 → 128, k=4, s=2, p=1	(128, 16, 16)
BatchNorm + LeakyReLU	—	—
Conv2d	128 → 256, k=4, s=2, p=1	(256, 8, 8)
BatchNorm + LeakyReLU	—	—
Conv2d	256 → 512, k=4, s=2, p=1	(512, 4, 4)
BatchNorm + LeakyReLU	—	—
Flatten	—	(8192)
Linear	8192 → 1	(1)

MODELING APPROACHES

DDPM E-UNET

1. Time Embedding

Sinusoidal embedding → MLP → 256-dim

2. Encoder (Downsampling Path)

Stage	Details	Output
Enc-1	Conv2d 3 → 64	(64, 64, 64)
ResBlock	64 → 64 (time-conditioned)	(64, 64, 64)
Enc-2	ResBlock 64 → 128	(128, 64, 64)
Downsample	stride=2	(128, 32, 32)
Enc-3	ResBlock 128 → 256	(256, 32, 32)
Downsample	stride=2	**(256, 16, 16)**

3. BOTTLENECK

Stage	Details	Output
ResBlock ×2	256 → 256	(256, 16, 16)

4. DECODER (UPSAMPLING PATH)

Stage	Details	Output
Up-1	Upsample + concat(skip)	(512, 32, 32)
ResBlock	512 → 128	(128, 32, 32)
Up-2	Upsample + concat(skip)	(256, 64, 64)
ResBlock	256 → 64	(64, 64, 64)
Final ResBlock	64 → 64	(64, 64, 64)
Output Layer	GN + SiLU + Conv2d 64 → 3	**(3, 64, 64)**

EVALUATION METRICS

FRÉCHET INCEPTION DISTANCE (FID)

- Measures how similar generated images are to real images
- Uses features extracted from InceptionV3
- Lower FID = better quality + more realistic images
- Captures:
  - Sharpness
  - Distribution similarity
  - Diversity

INCEPTION SCORE (IS)

- Uses Inception model to classify generated images
- Measures:
  - Image clarity (how confident the classifier is)
  - Diversity (variety of outputs)
- Higher IS = better

DCGAN

```
FID: 73.63
Inception Score: 1.85 ± 0.13
```

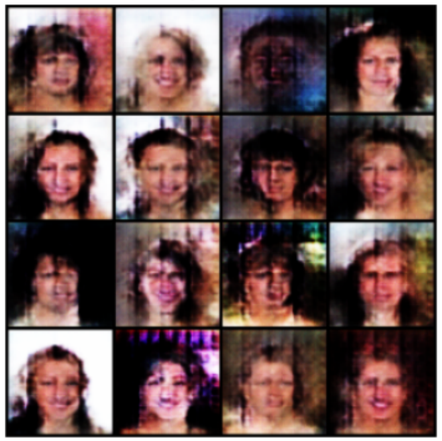
DIFFUSION MODEL

```
FID: 56.79
Inception Score: 1.90 ± 0.14
```

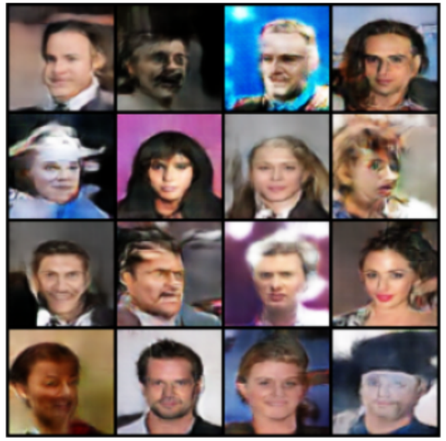
VISUAL RESULTS

DCGAN: Good global structure but blurry details, occasional artifacts

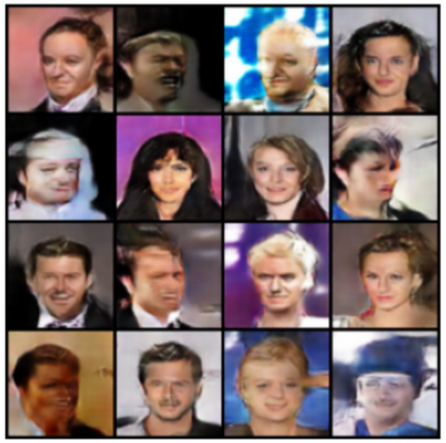
Epoch 1



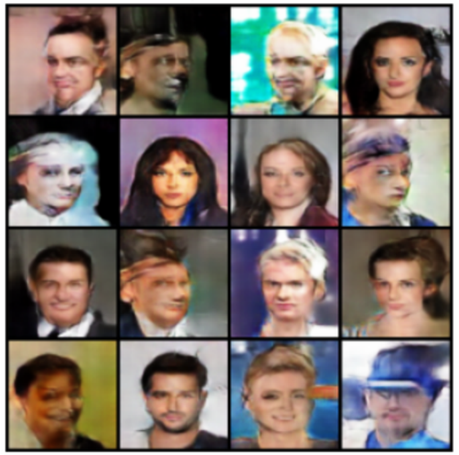
Epoch 20



Epoch 55



Epoch 60



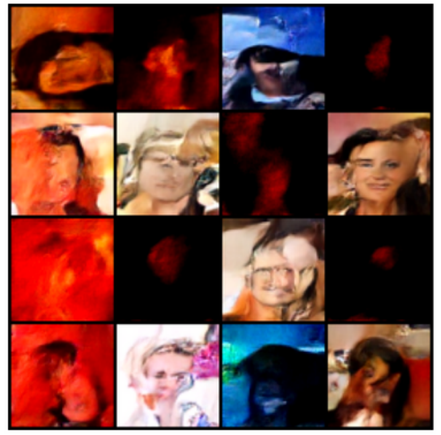
Epoch 100



VISUAL RESULTS

Diffusion: Sharper, more coherent facial features, better texture consistency

EPOCH1



EPOCH20



EPOCH55



EPOCH80



DISCUSSION

PERFORMANCE

- Lower FID (56.79 vs 73.63) → diffusion generated images closer to the real CelebA distribution
- Higher IS (1.90 vs 1.85) → slightly better clarity and diversity

TRAINING STABILITY

- DCGAN showed fluctuations in G-loss and D-loss

=> typical issues: instability, mode collapse, sensitivity to hyperparameters

- Diffusion training was more stable

=> due to the step-by-step denoising process, not adversarial competition

Computational Trade-offs

- Diffusion → better quality but slower sampling (multiple steps)
- DCGAN → faster generation but lower realism

VISUAL QUALITY DIFFERENCES

- DCGAN images:

=> recognizable faces but blurrier, some artifacts

- Diffusion images:

=> sharper, better textures, more coherent shapes (eyes, mouth, hair)

CONCLUSION

This study compared DCGAN and a Diffusion Model for face image synthesis on the CelebA dataset. While DCGAN produced acceptable images, it showed instability and lower visual quality. The Diffusion Model generated sharper, more realistic outputs and achieved better FID and IS scores, demonstrating greater stability and fidelity. Overall, diffusion models outperform DCGAN for high-quality image generation.

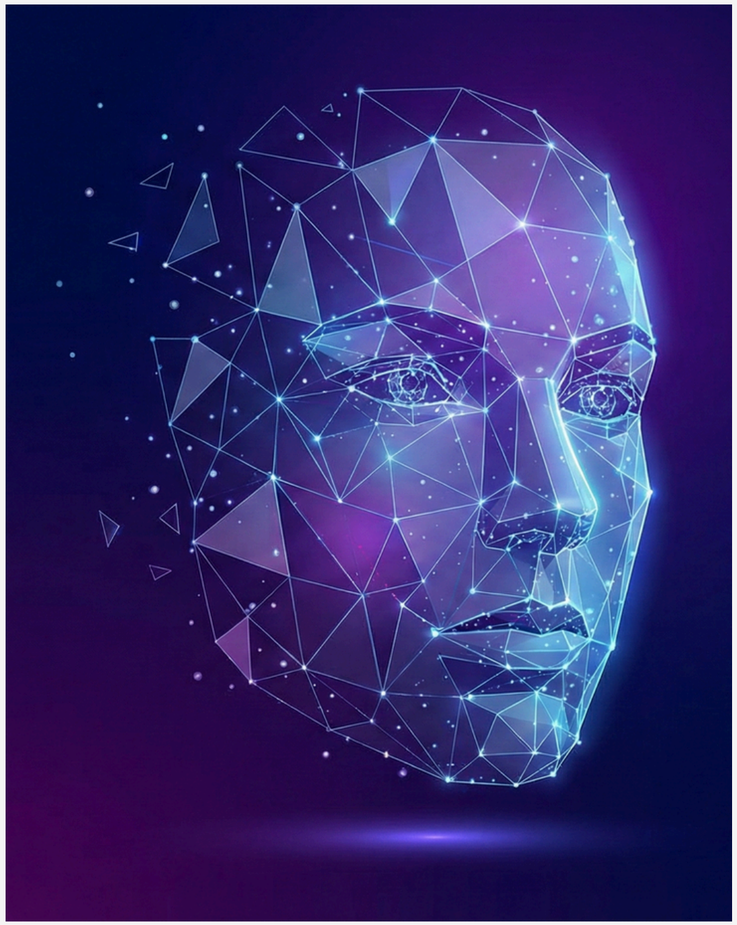
REFERENCES

[1] S. Chaturvedula and Y. Battineedi,
“Comparative Analysis of GANs and VAEs in Generating High-Quality Images: A Case Study on the MNIST Dataset,”
International Research Journal of Engineering and Technology (IRJET), vol. 11, no. 1, pp. 485–487, Jan. 2024.

[2] Z. Shen, M. Mao, and P. Fan,
“A Primary Comparison of Diffusion Models and Generative Adversarial Networks for Image Synthesis,”
Proceedings of the 7th International Conference on Machine Learning and Machine Intelligence (MLMI 2024), Osaka, Japan, 2024.
3696271.3696307

[3] L. N. Sequeira, P. L. de Paula Filho, E. Eyng, F. C. de Oliveira, and J. S. Andrade,
“Comparison Between GANs and Diffusion Models in the Generation of Synthetic Images for Enhancing Tree Species Recognition,”
Revista de Gestão Social e Ambiental, vol. 18, no. 10, pp. 1–16, 2024.
Comparison_Between_Gans_and_Dif...

[4] Y. Liu, H. Li, Z. Wang, and J. Zhang,
“Diffusion-Driven GAN Inversion for Multi-Modal Face Image Generation,”
arXiv preprint, 2024.



THANK YOU!